

Contents

Cross-species prediction of high-confidence human drug-induced liver injury from rat toxicogenomics: a leakage-safe TG-GATEs model and its external validation on DrugMatrix	1
Abstract	1
1. Introduction	2
2. Methods	3
2.1 Human DILI labels	3
2.2 Expression features (GSVA Hallmark)	3
2.3 Rat endpoint features	3
2.4 Chemistry features	3
2.5 Models and validation	3
2.6 External DrugMatrix pipeline	4
3. Results	4
3.1 Leakage-safe TG-GATEs model	4
3.2 Feature ablation (nested LOOCV, high-conf labels)	4
3.3 Where the signal lives (auxiliary analyses)	5
3.4 MCC-optimised fusion	5
3.5 External validation on DrugMatrix	5
3.6 The GSVA expression signature is the transferable model	6
3.7 Resolving the 7-negative limitation: reliable-negative mining and a merged cohort	7
4. Discussion	8
5. Limitations	8
6. Conclusion	9
Data and code availability	9
References (selected)	9

Cross-species prediction of high-confidence human drug-induced liver injury from rat toxicogenomics: a leakage-safe TG-GATEs model and its external validation on DrugMatrix

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Abstract

Background. Drug-induced liver injury (DILI) is a leading cause of drug attrition and post-market withdrawal. Rat toxicogenomic resources such as Open TG-GATEs pair *in vivo* liver transcriptomics with classical toxicology endpoints, but whether models trained on one such resource transfer to another — and which data modality carries the transferable signal — is rarely tested.

Methods. We predicted **high-confidence human DILI** labels (independent-Bayes fusion of seven bulk human-clinical sources; 101 drugs, 94 toxic / 7 safe) from Open TG-GATEs rat liver data using two feature modalities: (i) **pathway expression** — MSigDB Hallmark gene-set variation analysis (GSVA) scores derived from High-vs-Control rat log2 fold-changes mapped to human

orthologs; and (ii) **rat toxicology endpoints** — a curated pathology/chemistry rule (“OR-bit”: fatty change OR moderate-severe pathology OR relative-liver-weight increase). Models were evaluated by strictly **nested leave-one-drug-out cross-validation (LOOCV)**. We then rebuilt the entire feature pipeline independently, validated it against the frozen reference matrix, and applied the model to an external cohort from **DrugMatrix** (GEO GSE57815) — a different laboratory, platform batch, and drug set — reconstructing both feature arms from primary DrugMatrix expression and measured liver endpoints.

Results. The fused model (GSVA top-10 + OR-bit) reached **AUROC 0.790** (nested LOOCV; specificity 0.571 and MCC +0.462 at 95% sensitivity). A feature ablation showed the curated rat OR-bit was the single most predictive feature (AUROC 0.780), expression second (0.68–0.75), and chemistry weakest (0.52); only the parsimonious expression+OR fusion improved on either arm alone, while richer feature combinations overfit the seven negatives. An MCC-optimised two-feature fusion (expression signature + rat-panel probability) reached nested MCC +0.344 by exploiting complementary errors of the two arms. On the external **DrugMatrix** cohort (57 drugs, 47 toxic / 10 safe), the frozen expression arm **transferred without retraining** (GSVA signature AUROC 0.748 internal $\rightarrow$ 0.721 external, a -0.027 drop across lab/platform), even **out-predicting a model trained natively on DrugMatrix** (within-dataset LOOCV 0.632); whereas the frozen rat-endpoint arm **collapsed to chance** (0.47) despite the rat signal being genuinely present (DrugMatrix-native ALT $\rightarrow$ DILI AUROC 0.74) — a domain shift attributable to differing endpoint base rates (positive rate 0.68 vs 0.51). **Recalibrating the rat arm on DrugMatrix** and re-fusing with the frozen expression arm recovered **AUROC 0.791 / MCC +0.478**, matching internal performance on independent data.

Conclusions. Cross-species rat $\rightarrow$ human DILI signal replicates on an independent database. Pathway-level expression features are **portable across platforms**; measured rat-endpoint features are **more powerful but dataset-bound**, requiring per-dataset recalibration. We recommend deploying the expression arm as-is and refitting endpoint thresholds on each target dataset. Finally, reliable-negative mining and a merged TG-GATEs + DrugMatrix cohort lift the negative class from 7 to 36 and give a cross-platform model (AUROC 0.764, MCC +0.357) that validates within each platform separately — turning the fragile, small-negative operating point into a significant cross-dataset result.

1. Introduction

DILI is idiosyncratic, poorly predicted by standard preclinical assays, and a major reason drugs fail. Toxicogenomics — measuring transcriptomic responses to compounds in animal tissue — offers a mechanistic complement to apical toxicity endpoints. Two large public rat-liver resources dominate the field: **Open TG-GATEs** (NIBIOHN/NIHS) and **DrugMatrix** (Iconix, now hosted by the U.S. NTP), each pairing microarray expression with histopathology and clinical chemistry across hundreds of compounds.

A recurrent question is whether a rat toxicogenomic model predicts *human* clinical DILI, and whether such a model generalizes beyond the single resource it was trained on. Prior cross-species work often reports optimistic within-dataset cross-validation without an external test, and rarely disentangles which modality (expression vs measured phenotype vs chemistry) drives — and transfers — the signal.

Here we (1) build a leakage-safe rat→human DILI predictor on TG-GATEs against high-confidence human labels; (2) dissect the contribution and interaction of expression, rat endpoints, and chemistry; and (3) perform a genuine external validation on DrugMatrix, rebuilding both feature arms from primary DrugMatrix data. Our central finding concerns *differential transferability*: expression is portable, measured endpoints are potent but dataset-specific.

2. Methods

2.1 Human DILI labels

High-confidence binary labels were derived by independent-Bayes fusion of **seven bulk human-clinical sources** (DILIRank, DILIST, LiverTox, SIDER hepatic ADRs, Greene *humans* column, T3DB-clinical, ATSDR-clinical), excluding a hand-curated “Manual” source. A drug was retained if `n_sources_used` = 3 and the fused probability was extreme (`p_combined` = 0.80 or 0.20). This yielded **101 drugs (94 toxic / 7 safe)** with both TG-GATEs features and labels. The class imbalance (7 negatives) is intrinsic: the high-confidence filter is conservative and negatives are scarce among well-studied drugs.

2.2 Expression features (GSVA Hallmark)

From Open TG-GATEs rat Affymetrix liver expression (RMA, BrainArray ENTREZG): per drug we computed **log2 fold-change** as the High-dose treated-group mean minus the time-matched Control mean (falling back to Middle/Low dose when High was unavailable). Rat Entrez IDs were mapped to human gene symbols via orthology (mygene); multi-probe symbols were averaged. Sample-wise **GSVA** (gseapy, Gaussian kernel CDF) was scored against the 50 **MSigDB Hallmark** gene sets, giving a 101×50 pathway-enrichment matrix.

To confirm faithfulness of this independently reimplemented pipeline, we correlated it against the project’s frozen reference GSVA matrix: **per-drug median $r = 0.949$, per-pathway median $r = 0.970$** (overall Pearson 0.947). The nested champion computed on the rebuilt matrix reproduced the reference AUROC (0.795 vs 0.790).

2.3 Rat endpoint features

Open TG-GATEs pathology and clinical-chemistry endpoints at High dose were encoded as binary flags. The curated **OR-bit** rule combined three: `path_vacuol_fatty_high` OR `path_mod_severe_high` OR `chem_LWrel_1.2x_high`. A broader 14-endpoint panel (necrosis, hypertrophy, infiltration, ALT/AST $2\times/3\times$, relative liver weight, composites) was also used.

2.4 Chemistry features

RDKit 2-D molecular descriptors (~ 200) were computed from curated SMILES for the same drugs, used as a comparison modality.

2.5 Models and validation

The reference **champion** is `score = z(GSVA_topk10) + ·OR`, fit by **nested LOOCV**: within each training fold the top-10 Hallmark pathways were selected by $|AUC - 0.5|$, sign-corrected, z-scored and averaged into an expression signature; was chosen on the training fold to maximise training AUROC. The held-out drug never entered pathway selection, z-statistics, or — avoiding

the stacking optimism of the two-stage “OOF-then-fuse” variant (which inflated AUROC to 0.845 on the same cohort).

For the feature ablation and MCC-optimised fusion we used uniform learners (top-k z-signature per modality; L2-logistic regression; histogram gradient-boosting), all under nested LOOCV. Operating-point metrics (specificity, MCC) were reported at 95% sensitivity or at an MCC-optimal threshold chosen inside each fold (nested). AUROC 95% confidence intervals were bootstrap (2,000 resamples).

2.6 External DrugMatrix pipeline

DrugMatrix-Affymetrix expression (GEO **GSE57815**, RMA/BrainArray) was obtained from the unified mirror. **Vehicle controls** were identified from the GSE57815 series-matrix sample annotations as dose-0 samples, matched to treated samples by **vehicle and time**; per-drug High-vs-control log2FC was computed and passed through the *identical* ortholog→GSVA pipeline (Section 2.2), giving a 200×48 DrugMatrix Hallmark matrix.

Measured DrugMatrix liver endpoints (alanine/aspartate aminotransferase; hepatocyte lipid-accumulation and necrosis severities) were extracted from the ToxCompl input matrix `probe_clinical_pathology.xlsx` (NTP CEBS), decoded with the `C_H_M_annotation.txt` dictionary. We used **measured values only**; the ToxCompl *completed* (imputed) histopathology exhibits a positivity bias (predicted severities never reach zero) and was excluded. Endpoints were mapped to the TG-GATEs 14-endpoint schema (ALT/AST fold-change vs a normal-rat reference baseline; histopathology severity thresholds on the 0–4 scale).

The **external high-confidence cohort** comprised drugs with DrugMatrix GSVA and high-conf labels but **not** present in the TG-GATEs training set (69 drugs, 55 toxic / 14 safe); 57 (47 / 10) additionally had a full measured rat panel. Three regimes were evaluated: (a) **frozen** — TG-GATEs-trained model applied unchanged; (b) **diagnostic** — DrugMatrix-native endpoints vs DILI; (c) **recalibrated** — expression arm frozen, rat arm refit on DrugMatrix’s own endpoint scale, fused and evaluated by LOOCV within the external cohort.

3. Results

3.1 Leakage-safe TG-GATEs model

The nested champion achieved **AUROC 0.790** (95% CI [0.58, 0.97]); at 95% sensitivity, specificity 0.571 (4/7 safe), precision 0.97, **MCC +0.462**. Expression-only (no OR) gave AUROC 0.746. The two-stage OOF-then-fuse variant reached 0.845 but is optimistic and not reported as the result.

3.2 Feature ablation (nested LOOCV, high-conf labels)

Feature set	AUROC
Rat OR-bit (curated 3 endpoints)	0.780
Expression (GSVA; champion-tuned)	0.746
Expression (GSVA; uniform top-k)	0.681
Chemistry (200 RDKit descriptors)	0.521
Rat panel (14 endpoints, generic top-k)	0.454
Expression + OR-bit (champion)	0.795
Expression + chemistry / + full panel / all	0.44–0.51

Feature set	AUROC
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Three points emerge. (i) The **curated rat OR-bit is the single most predictive feature** — the *in vivo* phenotype outperforms transcriptomics on its own. (ii) **Curation matters more than modality**: the same rat endpoints give 0.780 as the 3-bit rule but 0.454 as a generic top-k over 14 endpoints. (iii) **Only the parsimonious expression+OR fusion helps**; adding chemistry, the full panel, or meta-stacking degraded performance, because seven negatives cannot support model complexity.

3.3 Where the signal lives (auxiliary analyses)

- Chemistry could **not** reproduce the rat OR call (chem→OR AUROC 0.557 chance): the phenotype is not deducible from structure.
- Expression **could** predict rat endpoints (expr→OR AUROC 0.827), but an OR imputed from expression was **redundant** with expression for DILI (Expr + imputed-OR = Expr alone), whereas Expr + *measured* OR reached 0.795. The measured phenotype therefore carries DILI-relevant information present in neither structure nor transcriptome.

3.4 MCC-optimised fusion

Because MCC is threshold-sensitive and dominated by the seven negatives, we optimised the operating point. The two arms failed on *different* safe drugs (e.g., expression rescued a rat false-positive, and vice-versa; inter-arm correlation = 0.30). A compact two-feature histogram-gradient-boosting model over [expression signature, rat-panel probability] reached **nested MCC +0.344**, exceeding either arm alone (rat +0.297; expression = 0). Concatenating all raw features instead overfit (AUROC 0.19–0.49) — distillation of each modality to one calibrated score before fusion was essential.

3.5 External validation on DrugMatrix

The rebuilt DrugMatrix GSVA passed a biological sanity check (–naphthyl-isothiocyanate, a cholestatic hepatotoxin, showed bile-acid-metabolism as a top down-regulated pathway).

Regime	Model	AUROC	95% CI	best MCC
Frozen	Expression (GSVA sig)	0.723	[0.55, 0.88]	+0.353
Frozen	Rat panel	0.470	[0.32, 0.61]	+0.173
Frozen	Full GBM (GSVA + rat)	0.499	[0.32, 0.69]	+0.081
Diagnostic	DrugMatrix- native ALT → DILI	0.737	—	—
Diagnostic	DrugMatrix- native OR-rule → DILI	0.627	—	—
Recalibrated	Expression (frozen)	0.723	[0.54, 0.88]	+0.353

Regime	Model	AUROC	95% CI	best MCC
Recalibrated	Rat arm (refit on DM scale)	0.736	[0.58, 0.87]	+0.408
Recalibrated	Expression(frozen) + rat(recal)	0.791	[0.64, 0.92]	+0.478

Three findings: (i) **expression transfers frozen** (0.72, matching internal); (ii) the **rat signal is real and replicates** (DrugMatrix-native ALT $\rightarrow$ DILI 0.74) but the **frozen** rat panel does not transfer (0.47) — a domain shift, with OR-positive base rate 0.68 in TG-GATEs vs 0.51 in DrugMatrix; (iii) **recalibrating the rat arm recovers champion-level performance on independent data** (0.791 / +0.478), with 10 negatives rather than 7.

3.6 The GSVA expression signature is the transferable model

The differential transferability above is carried by a single, parsimonious component: the **GSVA top-10 pathway signature**. Isolating it, the expression signature is the only feature that is simultaneously (a) strong under internal nested LOOCV on TG-GATEs and (b) portable — frozen, without any refitting or exposure to DrugMatrix labels — to an independent database.

GSVA signature — the transferable model.

Metric	TG-GATEs (internal LOOCV, n=101)	DrugMatrix (frozen transfer, n=57)	drop
AUROC (threshold-free)	0.748 (perm-p .013)	0.721 (perm-p .014)	−0.027
best-MCC	+0.462 (at Sens 0.95 / Spec 0.57)	+0.361 (at Sens 0.57 / Spec 0.90)	−0.10
MCC @ Sens 0.95	+0.462 (Spec 0.57)	+0.183 (Spec 0.20)	−0.28

The signature retains ~96% of its internal AUROC across a change of laboratory and platform batch (Affymetrix RG230 DrugMatrix vs TG-GATEs), and both AUROC’s are significant against a label-permutation null. AUROC (threshold-free) is the primary summary; MCC is threshold-dependent and we report it two ways to avoid an operating-point mismatch. **best-MCC** takes the MCC-optimal cutoff on each set: internally this coincides with the 95%-sensitivity point (+0.462), but on DrugMatrix the optimum trades sensitivity for specificity (Sens 0.57 / Spec 0.90) to reach +0.361. Held to a **fixed high-sensitivity operating point** (Sens 0.95, the convention of §3.1/§3.5), the transferred MCC is +0.183 — the honest cost of freezing a threshold across a base-rate shift. These are the expression-arm quantities of §3.1–§3.5 recomputed on the internal-champion training cohort (cf. expression-only 0.746, §3.2; frozen-expression best-MCC 0.353, §3.5).

Transfer beats native training. Under the *same* within-dataset LOOCV protocol applied to each database, the GSVA signature scores 0.748 on TG-GATEs but only **0.632 on DrugMatrix** (n=57) — because DrugMatrix self-LOOCV must train each fold on ~56 drugs with 10 negatives, too few to learn a stable signature. The frozen TG-GATEs signature transferred to DrugMatrix (**0.721**) therefore **exceeds DrugMatrix’s own within-dataset LOOCV (0.632)**: a model that never saw a DrugMatrix label predicts DrugMatrix DILI better than one trained on DrugMatrix itself.

The cross-species knowledge learned on TG-GATEs is genuinely additive, not merely recoverable from the target resource. (The 48-feature L2-logistic is not a reliable transferable model — it collapses to chance under internal LOOCV, 0.506, owing to the 7-negative imbalance; the low-parameter GSVA signature is the consistent choice across every protocol.)

3.7 Resolving the 7-negative limitation: reliable-negative mining and a merged cohort

The internal cohort’s seven negatives make operating-point metrics fragile — the specificity 95% CI spans the entire [0, 1]. We addressed this by **mining reliable negatives** and then **merging TG-GATEs with DrugMatrix** (details in [docs/MERGED_COHORT.md](#)).

Anchored on the 101 high-conf labels, a drug from the broader 402-drug rat-expression pool was accepted as a reliable new negative only when **two independent votes agreed it was safe**: the 7-source label fusion (`p_combined` 0.40) and an expression model retrained each round on the growing anchor set. The self-training loop converged in three rounds (+15, +9, +1; 7 → 32 negatives; imbalance 13:1 → ~3:1); every mined drug is a pharmacologically plausible low-DILI compound (corticosteroids, antiemetics, topical antiseptics, aminoglycosides, digoxin, vitamin D), and three (betamethasone, nystatin, neomycin) were independently present — and labelled SAFE — in the DrugMatrix external cohort. On the expanded internal cohort (champion frozen on TG-GATEs; every score out-of-sample), the specificity CI collapsed from [0, 1] to informative and the MCC 95% CI moved to **exclude zero** (94T/31S: AUROC 0.840, MCC@Sens0.95 +0.639 [+0.33, +0.76]).

Pooling both datasets then places **both classes on both platforms** (TG 94T/7S; DM 55T/29S), so a merged model cannot reduce to “platform = label”:

Merged champion (GSVA signature, LOOCV; 185 drugs, 149T/36S)	Value
AUROC	0.764 [0.66, 0.86]
MCC @ Sens 0.95	+0.357 [+0.13, +0.58]
within-TG-GATEs AUROC	0.767
within-DrugMatrix AUROC	0.746
platform-confound AUROC	0.620

The signal is DILI, not platform: the champion performs essentially identically within each platform (0.767 vs 0.746 vs pooled 0.764) — if the pooled result were platform-driven, the within-platform AUROCs would collapse to ~0.5 — and a predictor of *platform* reaches only 0.62, well below the DILI-label 0.764. The MCC is now significant on 36 negatives across two independent datasets, versus a non-significant MCC on 7. (Caveat: the mined negatives were expression-selected and expression-scored, so the *stability/significance* gains are fully valid but the *point-estimate lift* carries mild selection optimism; the within-TG-GATEs result, whose seven negatives are the untouched anchors, is the cleanest reference.)

Batch-correction sensitivity analysis (ComBat). Because the negatives are DrugMatrix-enriched, platform partly tracks the label; we therefore re-ran the merged champion after empirical-Bayes batch correction (ComBat, run unsupervised — no DILI-label covariate — so the correction cannot leak the outcome).

Merged champion	Uncorrected (GSVA-native)	ComBat-corrected
AUROC	0.764 [0.66, 0.86]	0.720 [0.62, 0.82]
MCC @ Sens 0.95	+0.357 [+0.13, +0.58]	+0.299 [+0.12, +0.48]
best-MCC	+0.460	+0.357
within-TG-GATEs AUROC	0.767	0.767
within-DrugMatrix AUROC	0.746	0.732
platform-confound AUROC	0.620	0.465

ComBat removed the platform structure (platform-confound AUROC $0.620 \rightarrow 0.465$ chance). The pooled AUROC fell modestly ($0.764 \rightarrow 0.720$) — the between-platform component that was partly confounded with the label — but the **within-platform AUROCs were essentially unchanged** (within-TG-GATEs identical at 0.767; within-DrugMatrix $0.746 \rightarrow 0.732$), and the MCC remained significant ($+0.299$ [$+0.12, +0.48$]). The merged model therefore *survives* explicit batch correction: the conservative cross-platform estimate is AUROC 0.720 / MCC $+0.299$, and the ~ 0.04 AUROC difference is attributable to (and removed as) platform effect rather than lost biology.

4. Discussion

The internal ranking (rat endpoints $>$ expression) *inverts* on transfer: the potent-but-parochial modality is the measured phenotype, whereas the portable modality is pathway-level expression. This is mechanistically sensible. Hallmark GSVA scores are relative, rank-based enrichments largely invariant to platform and batch, so a model frozen on them generalizes — indeed, the frozen TG-GATEs signature out-predicts a model trained natively on DrugMatrix (0.721 vs 0.632; §3.6), because rank-based pathway features carry a stable, transferable cross-species signal that the smaller target cohort cannot re-learn on its own. Apical endpoints, by contrast, depend on study-specific scoring conventions, dosing, and reference ranges; their marginal distributions differ across laboratories (base rate 0.68 vs 0.51), so a classifier frozen on one resource’s thresholds misfires on another’s — even though the underlying biology (ALT $\rightarrow$ DILI) is conserved (0.74).

Practically, this argues for a **hybrid deployment**: use the expression arm as a portable backbone, and refit endpoint thresholds/weights whenever a new dataset’s apical measurements are available. The complementarity of the two arms — retained cross-dataset — means the fusion is worth the recalibration cost.

Our auxiliary analyses further localise the DILI signal: it is **not** recoverable from chemistry alone, and it is **not** fully recoverable from transcriptome alone; the measured rat phenotype contributes orthogonal, irreducible information. This cautions against replacing *in vivo* endpoints with purely computational surrogates for this task.

5. Limitations

- **Small negative class.** Seven training and 10–14 external safe drugs make operating-point metrics (specificity, MCC) unstable and confidence intervals wide; AUROC (rank-based) is the more reliable summary. §3.7 mitigates this by mining reliable negatives and merging TG-GATEs with DrugMatrix (36 negatives, MCC significant), at the cost of the mild selection optimism noted there.
- **Recalibrated external results are LOOCV within a 57-drug cohort**, not a fully held-out third dataset; the expression arm is nonetheless frozen (true transfer), and the point

estimate matches internal performance on independent drugs.

- **Champion model selection** was made across candidate models on full-cohort LOOCV, introducing mild optimism; the nested numbers are the conservative reference.
- **Cross-platform batch effects** are absorbed at the pathway level but not eliminated; measured DrugMatrix histopathology coverage is sparser than clinical chemistry.

6. Conclusion

A leakage-safe rat→human DILI model fusing MSigDB-Hallmark expression with a curated rat toxicology rule reaches AUROC 0.790 on high-confidence human labels, and — critically — **generalizes to an independent database (DrugMatrix)** at champion-level performance (0.791 / MCC +0.478) once its rat arm is recalibrated. Expression features are portable across platforms; measured endpoint features are more powerful but dataset-bound. The measured *in vivo* phenotype carries DILI-relevant signal absent from both chemistry and transcriptome, underscoring the continued value of apical toxicogenomic endpoints. Finally, reliable-negative mining and a merged TG-GATEs + DrugMatrix cohort raise the negative class from 7 to 36 and yield a cross-platform model (AUROC 0.764, MCC +0.357) that validates *within each platform separately*, converting the fragile 7-negative operating point into a significant, cross-dataset result.

Data and code availability

Code (feature pipelines, nested-CV models, external-validation scripts) and derived artifacts (DrugMatrix GSVA matrix, measured endpoint panel, endpoint dictionary, per-drug scores, metrics) are in this repository, primarily under `scripts/external_validation/`, `data/external/`, and `results/external_validation/`; see `docs/EXTERNAL_VALIDATION_DRUGMATRIX.md`. The reliable-negative mining and merged-cohort champion (§3.7) are under `scripts/reliable_negatives/` with mined negatives committed at `data/reliable_negatives/round{1,2,3}.csv`; see `docs/MERGED_COHORT.md` (python `scripts/reliable_negatives/merged_champion.py`). Primary sources: Open TG-GATEs (NIBIOHN); DrugMatrix GSE57815 (GEO) and the unified rat-liver expression mirror (Hugging Face `ajaygeetakumar/dili-toxicogenomics-expression`); measured DrugMatrix endpoints and dictionary from NTP CEBS (ToxCompl); high-confidence human labels from the `entropy-or-highconf-human-dili` project.

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Prepared with computational assistance from Claude Code (Anthropic). Numerical results are from the analyses recorded in this repository.